

Luke Liang

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SUMMARY

Expertise in software engineering and ML applications. Nationally recognized for research in ML and simulations with *four published papers* as an undergraduate. Proven success in software engineering via an internship at JPMorgan Chase by *modernizing tools* and *leading a hackathon-winning project*. Developed *strong leadership skills* through extracurriculars. Passionate about applying technical solutions to solve complex problems.

SKILLS

Python, AI/ML, NumPy, TensorFlow, Network Science, Javascript, Numba, pandas, C++, ReactJS, Java, Linux, Git, HTML/CSS, MySQL, HPC, Docker, R, SAS

PROFESSIONAL EXPERIENCE

Teaching Assistant (ML & Algorithms)

Aug 2022 - Dec 2023

Miami University

Oxford, OH

- Led and published a final research project that evaluated the performance of several ML models and data compression techniques (**Python, ML**).
- Tutored students in Python, data structures, algorithms, and ML.
- Provided feedback and graded exams, homeworks, quizzes, and projects for students.

Software Engineering Intern

Jun 2022 - Aug 2022

JPMorgan Chase & Co

Chicago, IL

- Converted existing AngularJS internal software to ReactJS with a full-time team. Deployed the updated framework that is used throughout the entire bank (**ReactJS, Jenkins**).
- Won the Code for Good Hackathon 2022 with a web-based project to revamp a non-profit organization's website with a group of other interns (**Javascript, MySQL**).
- Integrated legacy pages with new internally developed APIs and updated technical documents.

Research Assistant

Aug 2021 - October 2024

Miami University

Oxford, OH

- Published 4 papers focused on ML and network science (*1 first author, 3 co-author*) ([Google Scholar](#)). Two first author manuscripts involving web applications and simulation are under preparation.
- Created a self-contained web-based project via Docker that houses a simulation platform for researchers at the CDC to evaluate policy effectiveness on youth suicide prevention (**Python, Numba, Docker, HPC**).
- Applied ML models and several methods to augment input data to differentiate between erroneous and correct simulation models (**Python, ML, TensorFlow, NumPy, pandas**).
- Developed a simulation of HIV drug resistance and mutation to evaluate treatment plans (**Python, NumPy**).
- Proposed a National Health Institute suicide modeling grant.

EDUCATION

BS in Computer Science, BS in Data Science, Math Minor (GPA: 3.86)

Aug 2021 - May 2024

Miami University

Oxford, OH

- Astronaut Scholarship (*national*), Choose Ohio First ML Scholarship, Summa Cum Laude, Honors College
- Received training at University of Tennessee (National Science Foundation sponsored) for ML and data science through the HPC at Oak Ridge national laboratory (*only student selected*).

LEADERSHIP EXPERIENCE

Chapter President

May 2023 - May 2024

Tau Beta Pi - The Engineering Honor Society

Oxford, OH

- Established the chapter as an official club through student government to receive funding for recruitment, social, and service events.
- Increased membership and community engagement by over 70% after inactivity due to COVID.